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Bifurcation Theory in Kaldor-Kalecki Business Cycle Model (macroeconomics)

Master degree in **Physics**

Course of **Mathematical Physics**

Padova, 06/2026

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Post Keynesian economics

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Classical and Neoclassical Economics
Late 1700s – early 1900s → Markets tend to self-correct

↓

Keynesian economics
1930s → Markets may fail at the macro level, central intervention may be needed

→ **Neoclassical Keynesianism**
1940s – 1970s → IS – LM framework

→ **Post-Keynesian economics**
1950s → Kaldor-Kalecki Business Cyle Model

Kaldor-Kalecki Business Cycle Model

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$$\begin{cases} \dot{Y} = \alpha\,[I(Y,K) - S(Y,K)] \\ \dot{K} = I(Y,K) - \delta\,K \end{cases}$$

$Y > 0$ (gross product) $\qquad K > 0$ (capital stock)

$\alpha > 0$ (control parameter) $\qquad \delta > 0$ (depreciation)

$I(Y,K)$: investment function $\qquad S(Y,K)$: savings function

$I > S \quad \Rightarrow \quad Y$ increases (boom)

$I < S \quad \Rightarrow \quad Y$ decreases (recession)

Poincaré – Bendixson Theorem

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Let $\dot{x} = f(x), x \in \mathbb{R}^2, f \in C^k$ $(k \geq 1)$ and let R be a compact (closed and bounded) region of $\mathbb{R}^2$. Suppose $\mathcal{C}$ is an orbit which is trapped in R. Then:

1. Either $\mathcal{C}$ is a closed orbit
2. Or it approaches
    1. A closed orbit $\longleftrightarrow$ $\alpha > \alpha_C$
    2. An equilibrium point $\longleftrightarrow$ $\alpha < \alpha_C$
    3. A cycle graph

Saving function

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Keynesian Economics:

$$Y = C + S\,,$$

where C is the consumption and S is the saving.

Assume

$$C = (1-\gamma)\,Y, \quad 0 < \gamma < 1 \qquad \Longrightarrow \qquad S = S(Y) = \gamma\,Y$$

Then

$$\begin{cases} \dot{Y} = \alpha\,[I(Y,K) - \gamma\,Y] \\ \dot{K} = I(Y,K) - \delta\,K \end{cases}$$

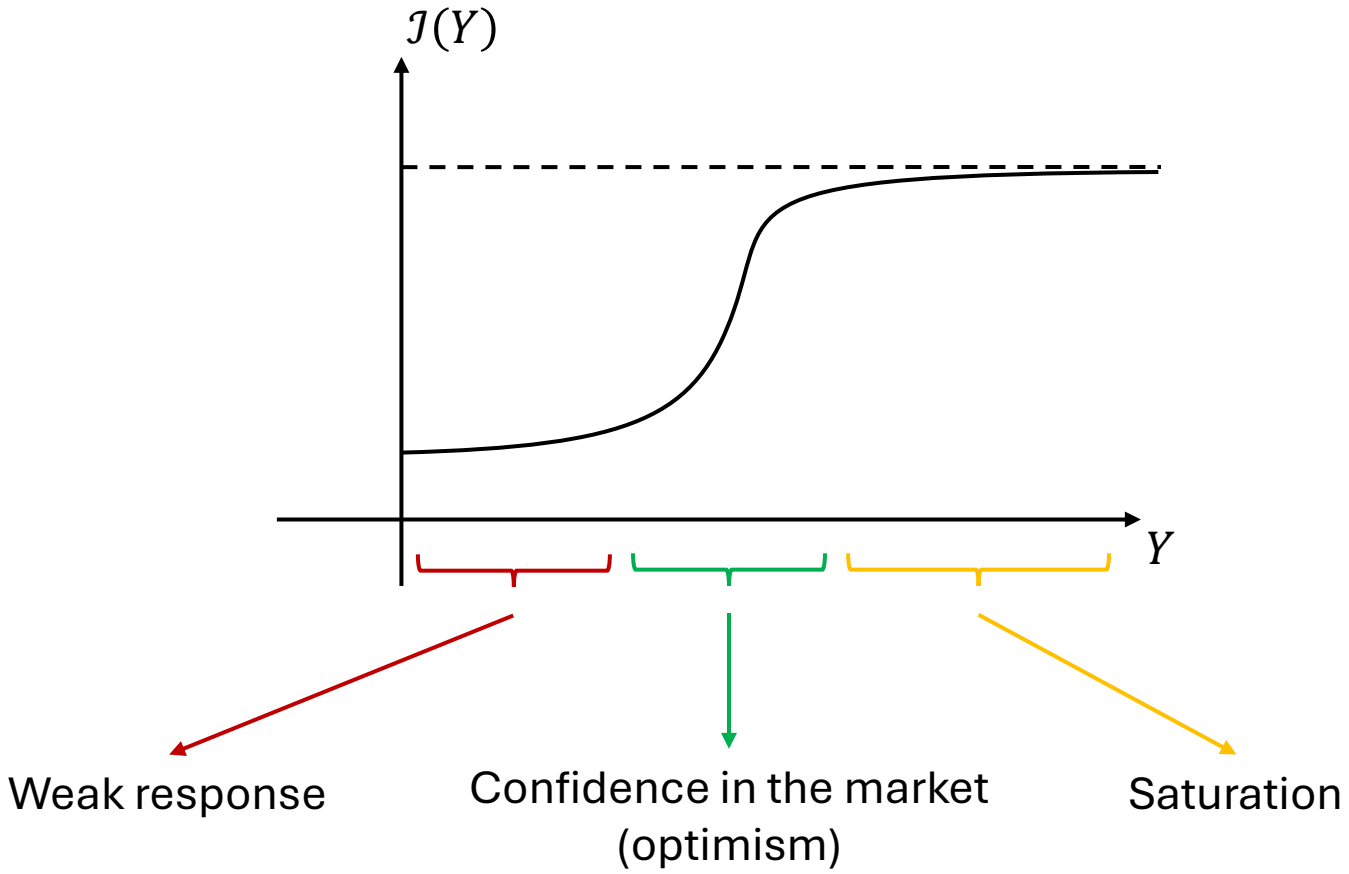
Investment function

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$I(Y,K) = \mathcal{I}(Y) - \beta\, K \qquad \beta > 0$

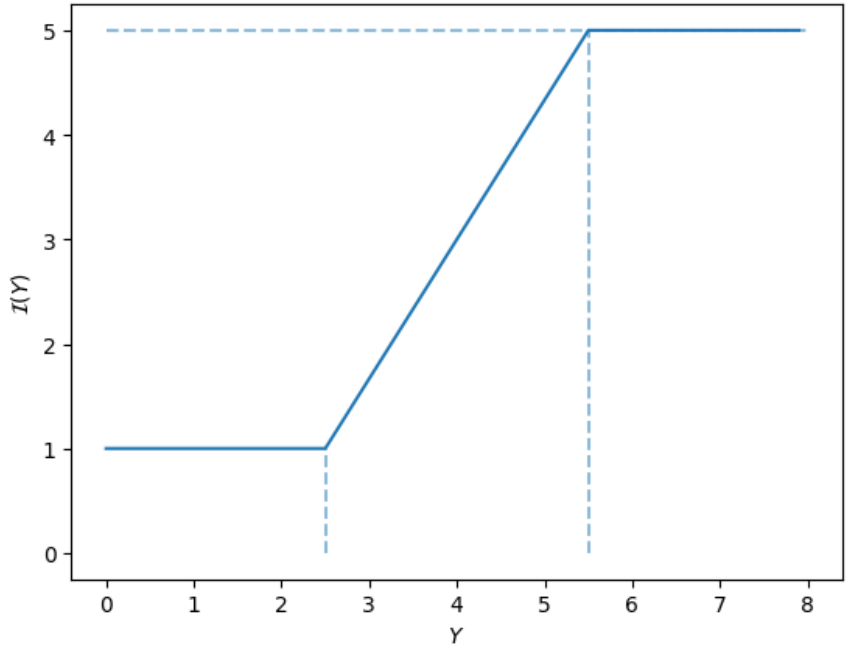
$\mathcal{I}(Y)$ s-shaped function

→ Trust in the markets



Piecewise investment function

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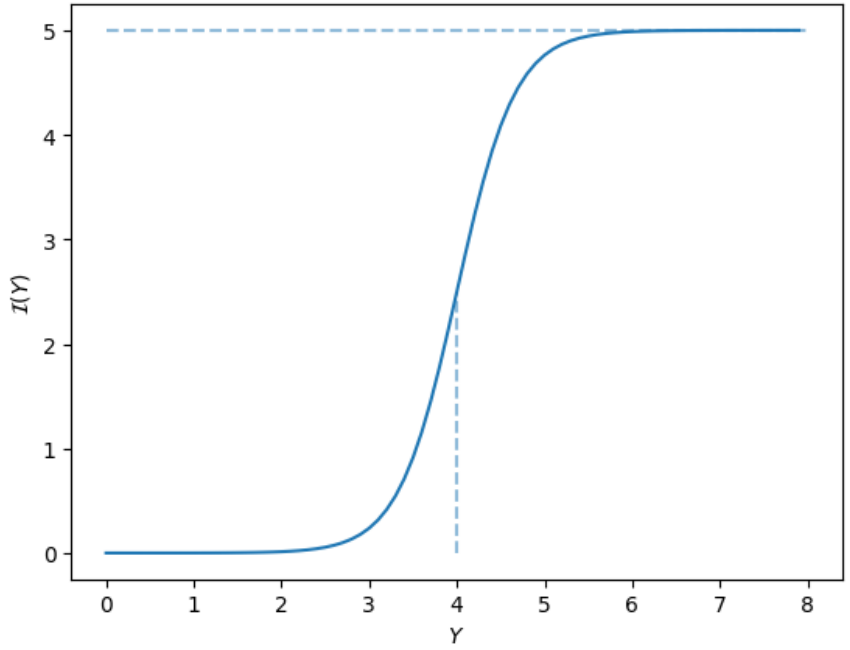
$$\mathcal{I}(Y) = \begin{cases} A & Y \leq Y_1 \\ A + \dfrac{B - A}{Y_2 - Y_1} Y & Y_1 < Y < Y_2 \\ B & Y \geq Y_2 \end{cases}$$

$A = 1 \quad B = 5 \quad Y_1 = 2.5 \quad Y_2 = 5.5$

- Simple (analytical result)
- not very realistic
- Non-smooth

Sigmoid investment function

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$$\mathcal{I}(Y) = \frac{A}{1 + e^{-B(Y-Y_0)}}$$

- Numerical result
- Smooth

$A = 5 \quad B = 3 \quad Y_0 = 4$

Kaldor-Kalecki Business Cycle Model

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$$\begin{cases} \dot{Y} = \alpha[\mathcal{I}(Y) - \beta\, K\ - \gamma\, Y] \\ \dot{K} = \mathcal{I}(Y) - (\beta + \delta)\, K \end{cases}$$

$$\alpha > 0, \qquad \beta > 0,$$
$$\delta > 0, \qquad 0 < \gamma < 1.$$

$$f(Y,K) = \begin{pmatrix} \alpha[\mathcal{I}(Y) - \beta\, K\ - \gamma\, Y] \\ \mathcal{I}(Y) - (\beta + \delta)\, K \end{pmatrix}$$

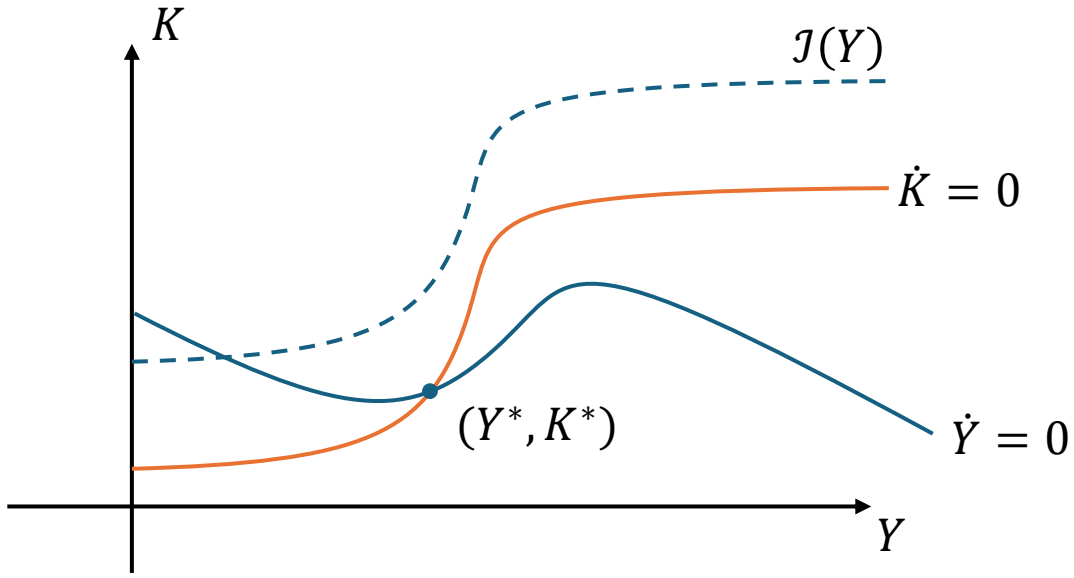
$$J_f(Y,K) = \begin{pmatrix} \alpha[\mathcal{I}'(Y) - \gamma\ ] & -\alpha\,\beta \\ \mathcal{I}'(Y) & -(\beta + \delta) \end{pmatrix}$$

Null-clines

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$$\dot{Y} = 0 \qquad \Leftrightarrow \qquad K = -\frac{\gamma}{\beta}\, Y + \frac{\mathcal{I}(Y)}{\beta}$$

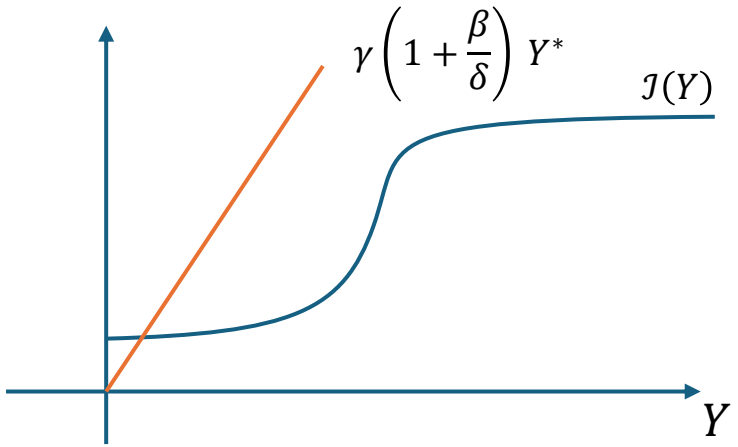
$$\dot{K} = 0 \qquad \Leftrightarrow \qquad K = \frac{\mathcal{I}(Y)}{\beta + \delta}$$



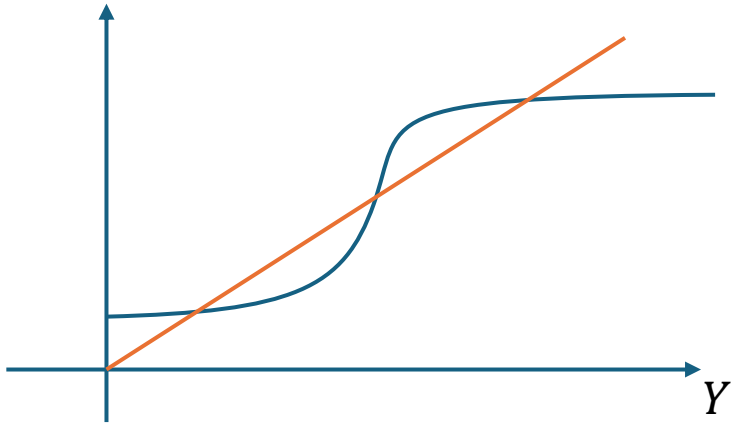
Equilibria

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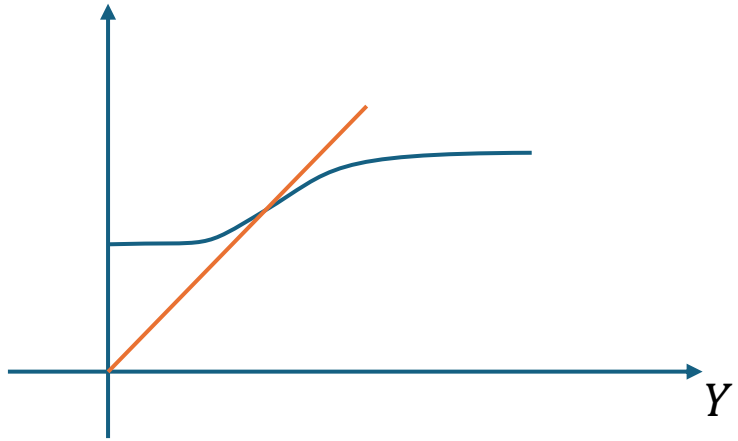
$$\begin{cases} \mathcal{I}(Y) - \beta K - \gamma Y = 0 \\ \mathcal{I}(Y) - (\beta + \delta) K = 0 \end{cases} \quad \Rightarrow \quad K^* = \frac{\gamma}{\delta} Y^*, \quad \mathcal{I}(Y^*) = \gamma \left(1 + \frac{\beta}{\delta}\right) Y^*$$



1 equilibrium in the pessimistic region



3 equilibria



1 equilibrium in the optimistic region

Choose an equilibrium point (Y^*, K^*)

$$A = J_f(Y^*, K^*) = \begin{pmatrix} \alpha[\mathcal{I}'(Y^*) - \gamma\,] & -\alpha\,\beta \\ \mathcal{I}'(Y^*) & -(\beta + \delta) \end{pmatrix}$$

$$\operatorname{tr} A = \alpha[\mathcal{I}'(Y^*) - \gamma] - (\beta + \delta)$$

$$\alpha_C \quad : \quad \operatorname{tr} A = 0$$

$$\alpha_C = \frac{\beta + \delta}{\mathcal{I}'(Y^*) - \gamma}$$

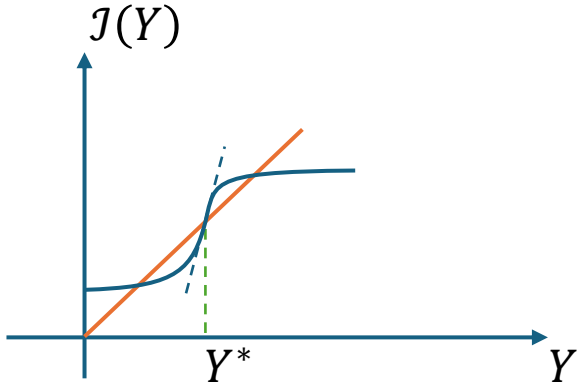
Eigenvalues

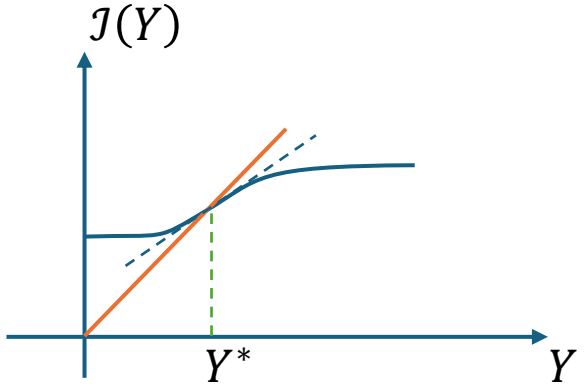
$$\det A = -\alpha[\delta\, \mathcal{I}'(Y^*) - \gamma(\beta + \delta)]$$

$$\lambda_\pm = \frac{1}{2}\left[\operatorname{tr} A \pm \sqrt{\Delta}\right] \qquad \Delta = (\operatorname{tr} A)^2 - 4 \det A$$

$$\alpha \simeq \alpha_C \qquad \Rightarrow \qquad \Delta \simeq -4 \det A$$

- $\lambda_\pm \in \mathbb{R}$ if $\det A < 0 \quad \Leftrightarrow \quad \mathcal{I}'(Y^*) > \gamma\left(1 + \frac{\beta}{\delta}\right)$
- $\lambda_\pm \in \mathbb{C}$ if $\det A > 0 \quad \Leftrightarrow \quad \mathcal{I}'(Y^*) < \gamma\left(1 + \frac{\beta}{\delta}\right)$





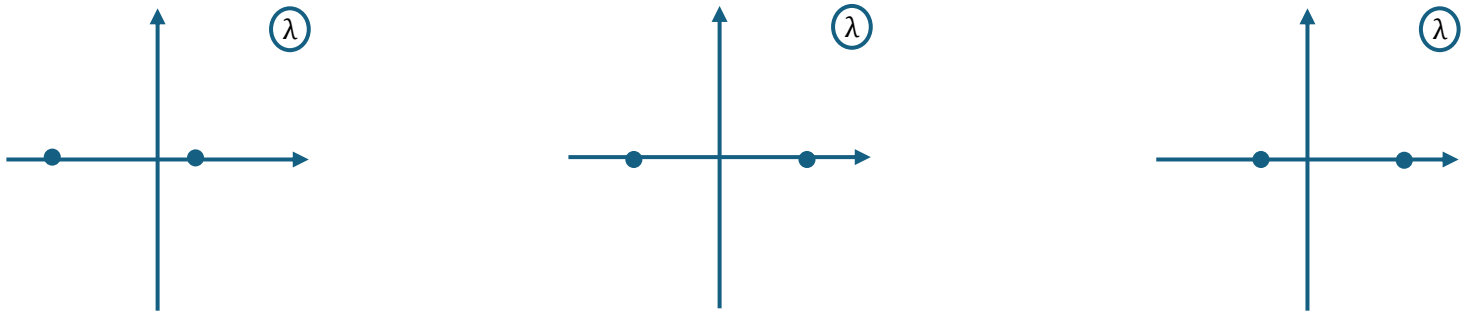
Real eigenvalues

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$$\lambda_{\pm} = \frac{1}{2}\left[\operatorname{tr} A \pm \sqrt{\Delta}\right] \qquad \Delta = (\operatorname{tr} A)^2 - 4 \det A$$

$$\sqrt{\Delta} > |\operatorname{tr} A| \qquad \Rightarrow \qquad \lambda_+ > 0, \quad \lambda_- < 0$$

$\alpha < \alpha_C$ $\alpha = \alpha_C$ $\alpha > \alpha_C$



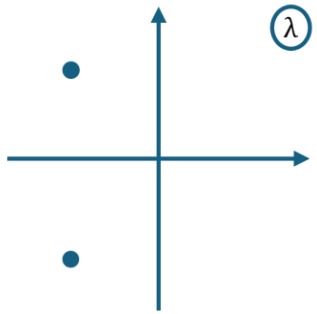
Complex eigenvalues

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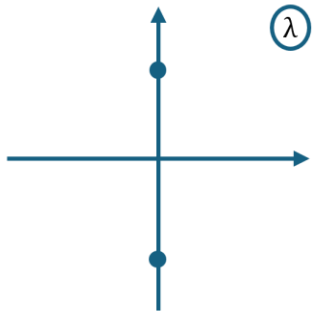
$$\lambda_{\pm} = \frac{1}{2}\left[\operatorname{tr} A \pm \sqrt{\Delta}\right] \qquad \Delta = (\operatorname{tr} A)^2 - 4 \det A$$

$$\operatorname{Re}(\lambda_{\pm}) = \frac{1}{2} \operatorname{tr} A$$

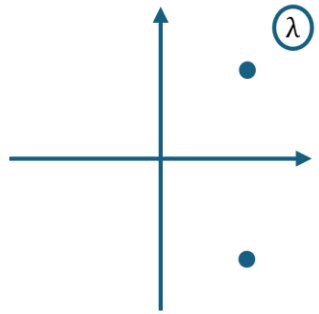
$\alpha < \alpha_C$



$\alpha = \alpha_C$

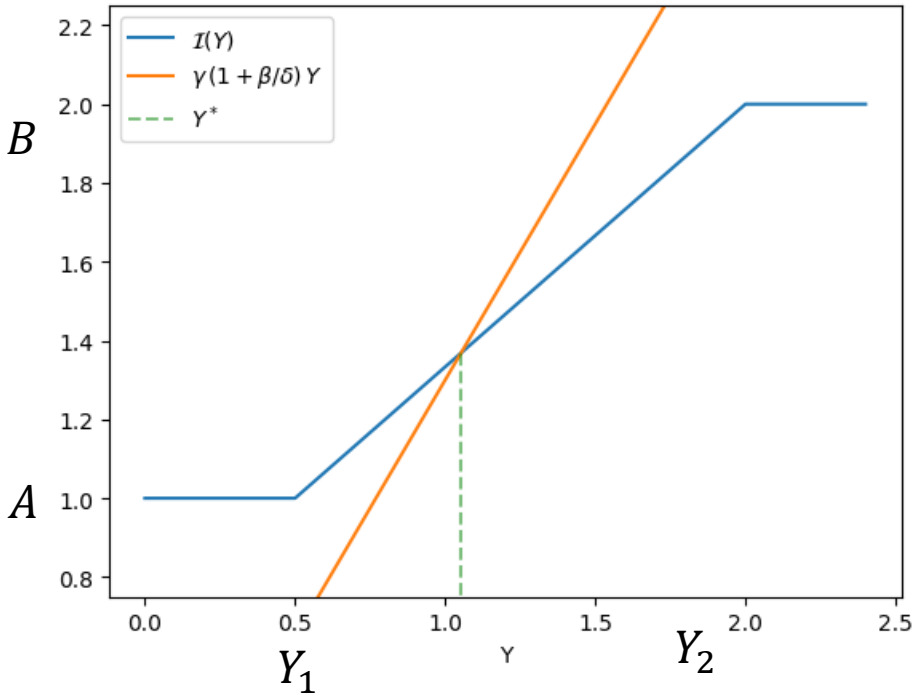


$\alpha > \alpha_C$



Piecewise investment function

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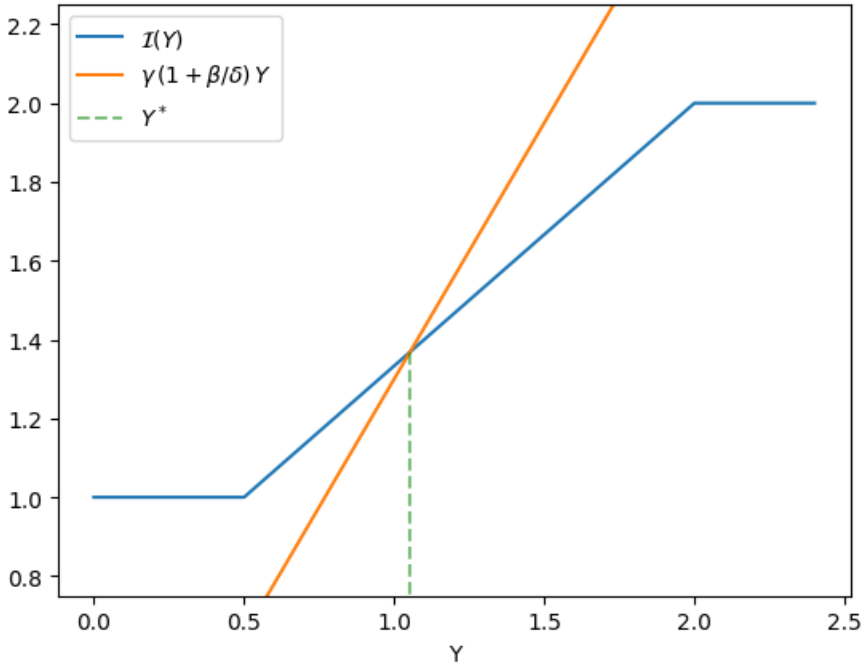
Condition: $Y_1 < Y^* < Y_2$

$$\mathcal{I}(Y^*) = \gamma\left(1+\frac{\beta}{\delta}\right) Y^* \qquad \Rightarrow$$

$$Y^* = \frac{\frac{(B-A)\,Y_1}{Y_2-Y_1} - A}{\frac{(B-A)}{Y_2-Y_1} - \gamma\left(1+\frac{\beta}{\delta}\right)}$$

$\lambda_{\pm} \in \mathbb{C}$

$$\mathcal{I}'(Y^*) = \frac{B-A}{Y_2-Y_1} < \gamma\left(1+\frac{\beta}{\delta}\right)$$



$$\beta = 0.8 \qquad A = 1$$
$$\gamma = 0.5 \qquad B = 2$$
$$\delta = 0.5 \qquad Y_1 = 0.5$$
$$Y_2 = 2$$

$$K^* = Y^* \simeq 1.053 \qquad (\gamma = \delta)$$

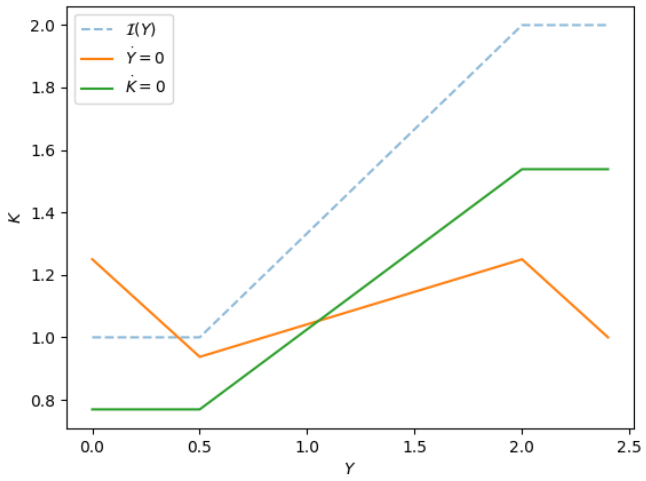
$$\alpha_C \simeq 7.8$$

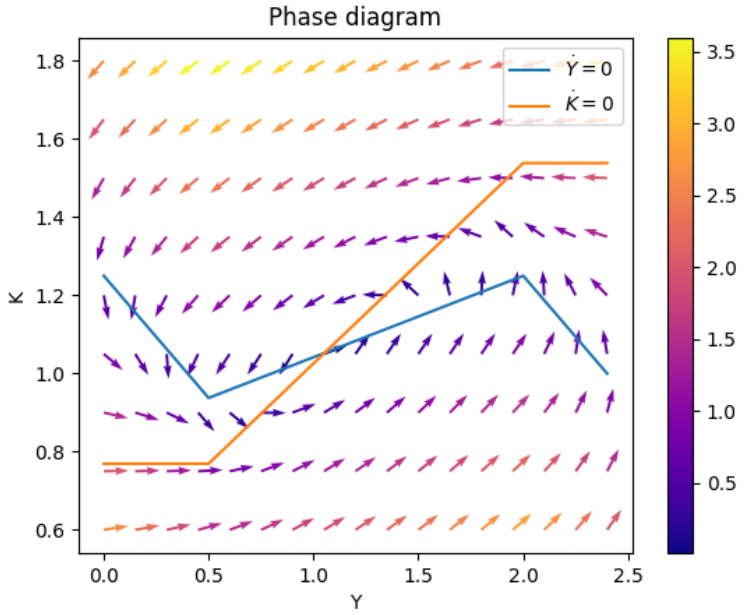
Null-clines and vector field

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$$\dot{Y} = 0 \quad \Leftrightarrow \quad K = -\frac{\gamma}{\beta} Y + \frac{\mathcal{I}(Y)}{\beta} \quad , \quad \dot{K} = 0 \quad \Leftrightarrow \quad K = \frac{\mathcal{I}(Y)}{\beta + \delta}$$

$$\alpha = 5$$

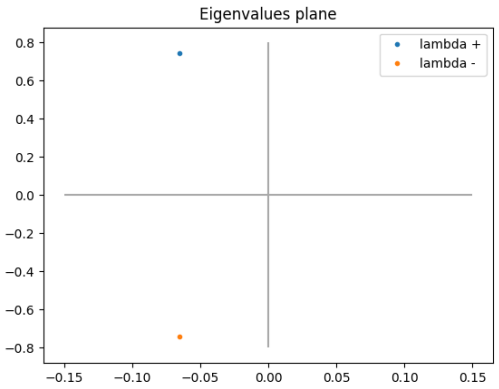


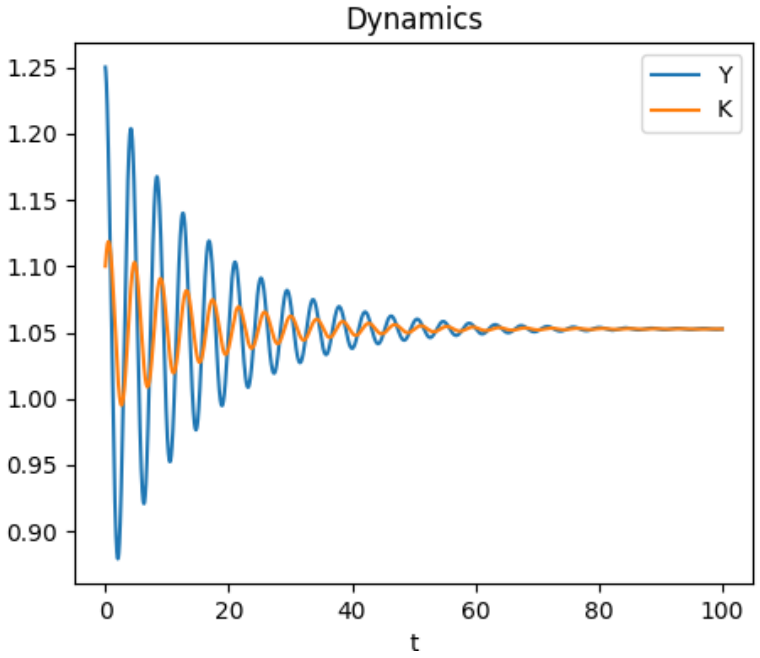


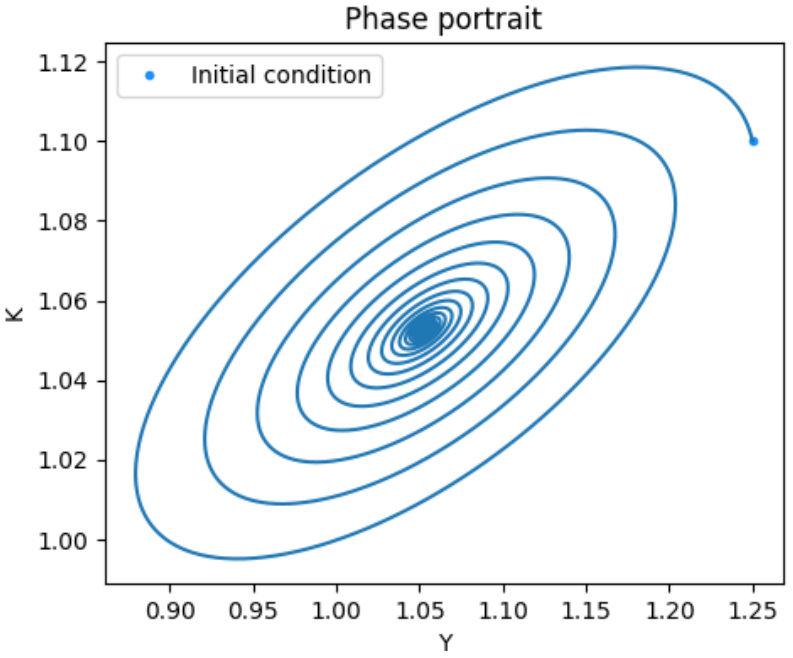
Numerical simulation

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$\alpha = 0.9\ \alpha_C$ numerical simulations using `odeint` (from `scipy`, python)



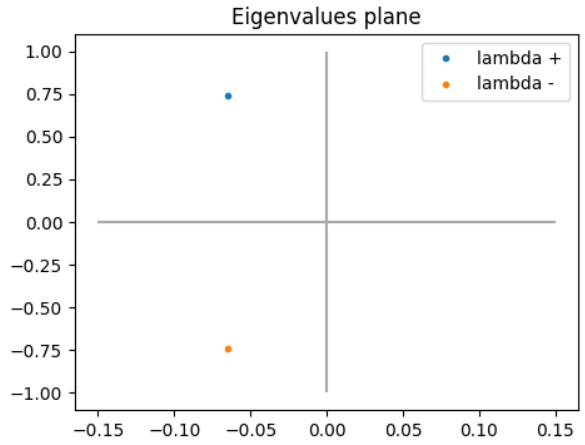


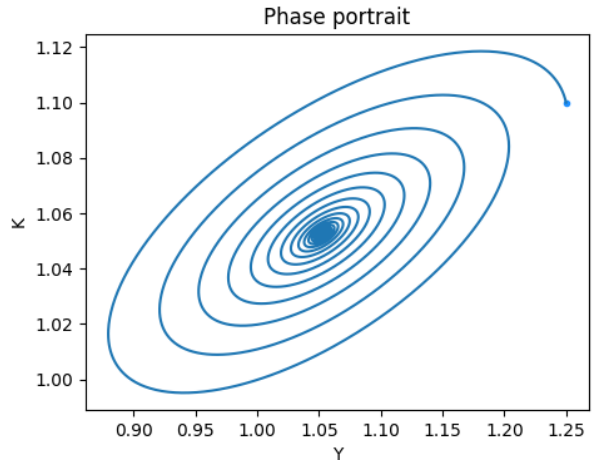


Bifurcation

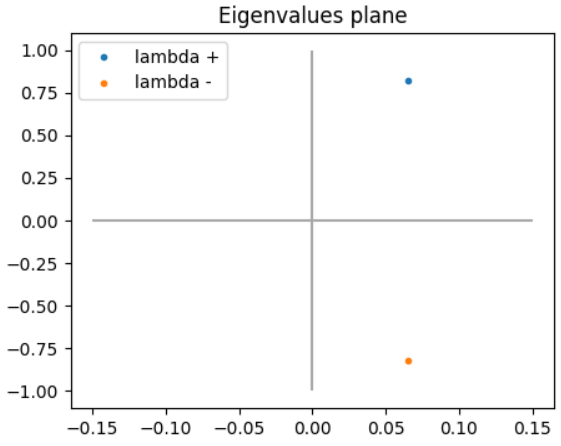
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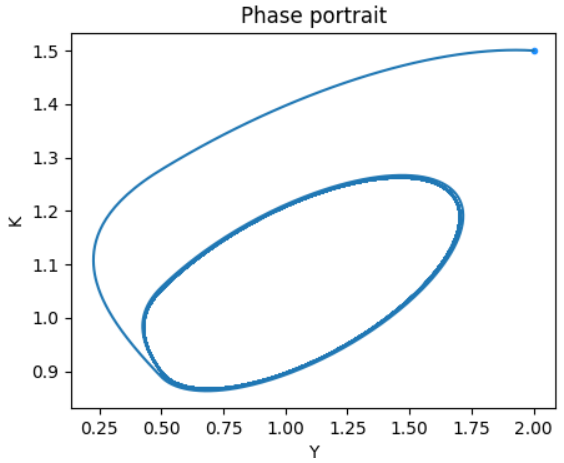
$\alpha = 0.9\,\alpha_C$

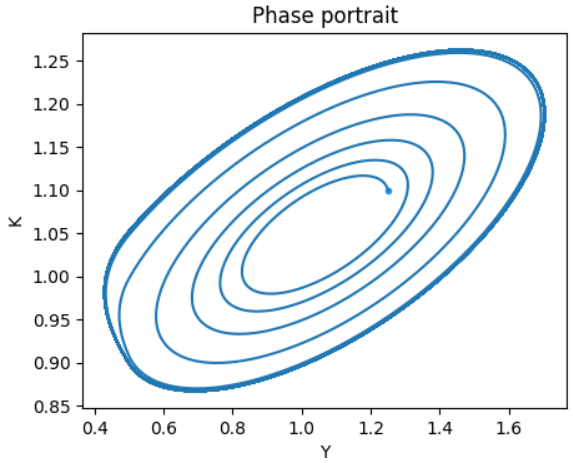




$\alpha = 1.1\,\alpha_C$

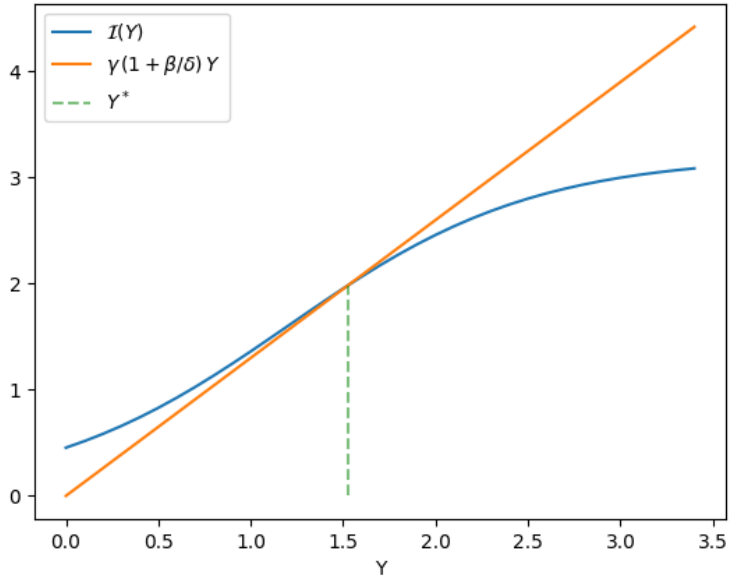






Sigmoid investment function

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$$\beta = 0.8 \qquad A = 3.2$$
$$\gamma = 0.5 \qquad B = 1.5$$
$$\delta = 0.5 \qquad Y_0 = 1.2$$

$$\frac{A}{1 + e^{-B(Y^* - Y_0)}} = \gamma \left(1 + \frac{\beta}{\delta}\right) Y^*$$

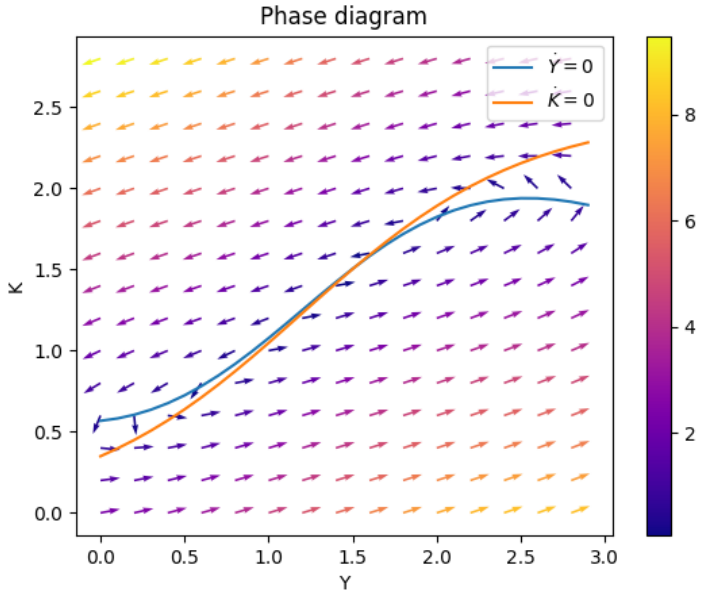
$$\Rightarrow \qquad Y^* = K^* \simeq 1.52 \qquad (\gamma = \delta)$$

Null-clines and vector field

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$$\dot{Y} = 0 \quad \Leftrightarrow \quad K = -\frac{\gamma}{\beta} Y + \frac{\mathcal{I}(Y)}{\beta} \quad , \quad \dot{K} = 0 \quad \Leftrightarrow \quad K = \frac{\mathcal{I}(Y)}{\beta + \delta}$$

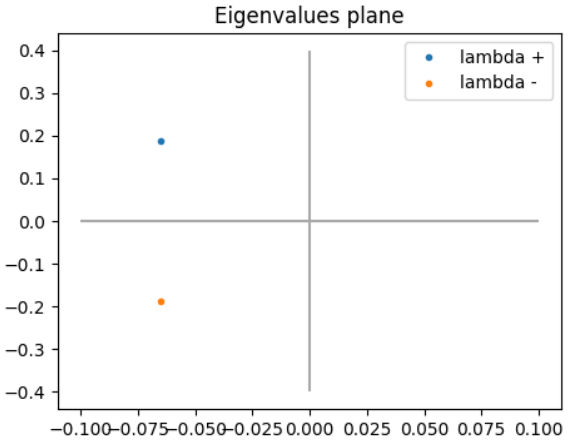
$$\alpha = 5$$

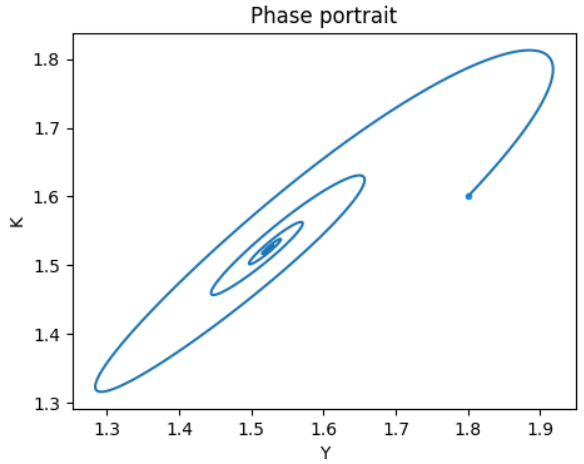


Bifurcation

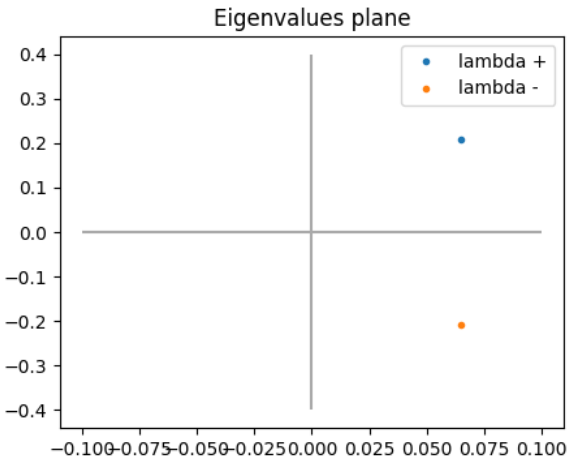
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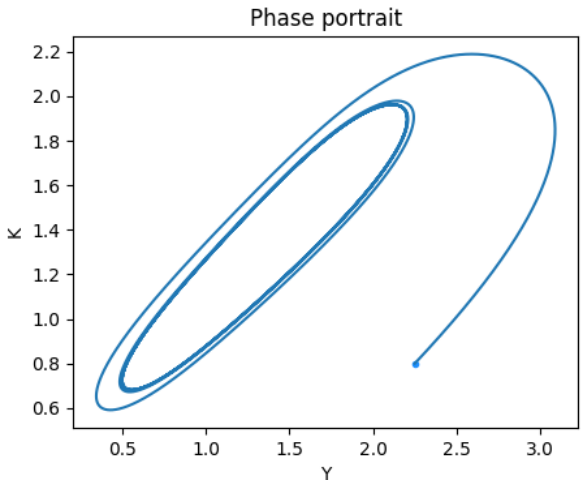
$\alpha = 0.9\,\alpha_C$

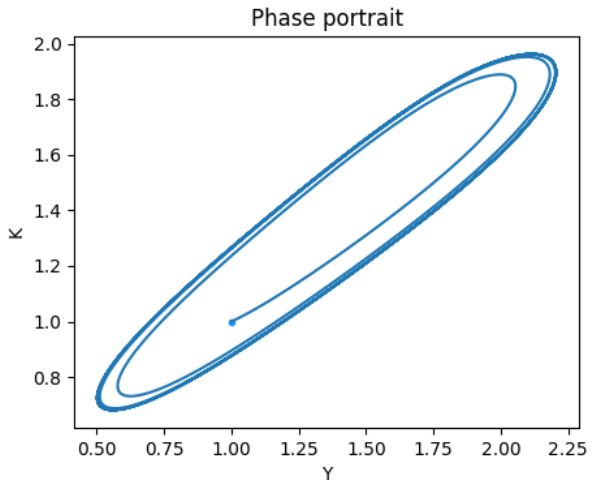




$\alpha = 1.1\,\alpha_C$







Translation

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$$\begin{cases} \dot{Y} = \alpha[\mathcal{I}(Y) - \beta\, K \ - \gamma\, Y] \\ \dot{K} = \mathcal{I}(Y) - (\beta + \delta)\, K \end{cases}$$

Choose an equilibrium point (Y^*, K^*)

$$\tilde{Y} = Y - Y^* \quad \tilde{K} = K - K^*$$

$$\begin{cases} \dot{\tilde{Y}} = \alpha[\mathcal{I}(\tilde{Y} + Y^*) - \mathcal{I}(Y^*) - \beta\, \tilde{K} \ - \gamma\, \tilde{Y}] \\ \dot{\tilde{K}} = \mathcal{I}(\tilde{Y} + Y^*) - \mathcal{I}(Y^*) - (\beta + \delta)\, \tilde{K} \end{cases}$$

Normal Form

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Taylor expansion:

$$\mathcal{I}\left(\tilde{Y}+Y^*\right)-\mathcal{I}(Y^*)=\underbrace{\frac{d\,\mathcal{I}}{d\,Y}(Y^*)}_{\mathcal{I}_0'}\tilde{Y}+\frac{1}{2}\underbrace{\frac{d^2\,\mathcal{I}}{d\,Y^2}(Y^*)}_{\mathcal{I}_0''}\tilde{Y}^2+\frac{1}{6}\underbrace{\frac{d^3\,\mathcal{I}}{d\,Y^3}(Y^*)}_{\mathcal{I}_0'''}\tilde{Y}^3+\mathcal{O}(\tilde{Y}^4)$$

$$\tilde{Y}\to Y \qquad \tilde{K}\to K$$

$$\begin{cases}\dot{Y}=\alpha\left[(\mathcal{I}_0'-\gamma)Y-\beta K+\dfrac{\mathcal{I}_0''}{2}Y^2+\dfrac{\mathcal{I}_0'''}{6}Y^3+\mathcal{O}(Y^4)\right]\\ \dot{K}=\mathcal{I}_0'Y-(\beta-\delta)K+\dfrac{1}{2}\mathcal{I}_0''Y^2+\dfrac{1}{6}\mathcal{I}_0'''Y^3+\mathcal{O}(Y^4)\end{cases}$$

$\lambda_{\pm} \in \mathbb{C}$

$$\lambda_{\pm} = \lambda_r \pm \lambda_c \boldsymbol{i}$$

$$\lambda_r = \frac{\operatorname{tr} A}{2} \qquad \lambda_c = \frac{\sqrt{4 \det A - (\operatorname{tr} A)^2}}{2}$$

$$\begin{cases} \dot{Y} = \lambda_r Y - \overbrace{[\alpha\beta\, K - (\lambda_r + \beta + \delta)\, Y]}^{\phi} + \alpha \left[\frac{\mathcal{I}_0''}{2} Y^2 + \frac{\mathcal{I}_0'''}{6} Y^3 + \mathcal{O}(Y^4) \right] \\ \dot{K} = \mathcal{I}_0' Y - (\beta - \delta) K + \frac{1}{2} \mathcal{I}_0'' Y^2 + \frac{1}{6} \mathcal{I}_0''' Y^3 + \mathcal{O}(Y^4) \end{cases}$$

$$\begin{cases} \phi = \alpha\beta\, K - (\lambda_r + \beta + \delta)\, Y \\ \varphi = \lambda_c\, Y \end{cases} \quad \Leftrightarrow \quad \begin{cases} Y = \dfrac{\varphi}{\lambda_c} \\ K = \dfrac{1}{\alpha\beta}\left[\phi + (\lambda_r + \beta + \delta)\, Y\right] \end{cases}$$

$$\begin{cases} \dot{\varphi} = \lambda_r\, \varphi - \lambda_c\, \phi + \dfrac{\alpha\, \mathcal{I}_0''}{2\, \lambda_c}\, \varphi^2 + \dfrac{\alpha\, \mathcal{I}_0'''}{6\, \lambda_c^2}\, \varphi^3 + \mathcal{O}(\varphi^4) \\ \dot{\phi} = \lambda_r\, \phi + \lambda_c\, \varphi - \dfrac{\alpha\, (\lambda_r + \delta)\, \mathcal{I}_0''}{2\, \lambda_c^2}\, \varphi^2 - \dfrac{\alpha\, (\lambda_r + \delta)\, \mathcal{I}_0'''}{6\, \lambda_c^3}\, \varphi^3 + \mathcal{O}(\varphi^4) \end{cases} \longrightarrow$$

Can we remove the quadratic and cubic term?

Denoting $\Lambda = \arg(\lambda)$

$$\begin{cases} \dot{\varphi} = |\lambda|\cos(2\pi\Lambda)\varphi - |\lambda|\sin(2\pi\Lambda)\,\phi + \dfrac{\alpha}{2\lambda_c}\mathcal{I}_0''\varphi^2 + \dfrac{\alpha}{6\lambda_c^2}\mathcal{I}_0'''\varphi^3 + \mathcal{O}(\varphi^4), \\ \dot{\phi} = |\lambda|\cos(2\pi\Lambda)\varphi + |\lambda|\sin(2\pi\Lambda)\,\phi - \dfrac{\alpha\delta}{2\lambda_c^2}\mathcal{I}_0''\varphi^2 + \dfrac{\alpha\delta}{6\lambda_c^3}\mathcal{I}_0'''\varphi^3 + \mathcal{O}(\varphi^4), \end{cases}$$

Define

$$z = \varphi + \phi\,\boldsymbol{i} \qquad \Leftrightarrow \qquad \begin{cases} \varphi = \dfrac{1}{2}(z + \bar{z}) \\ \phi = -\dfrac{\boldsymbol{i}}{2}(z - \bar{z}) \end{cases}$$

$$\begin{cases} \dot{z} = \lambda z + a_{20}z^2 + 2a_{11}z\bar{z} + a_{02}\bar{z}^2 + b_{30}z^3 + 3b_{21}z^2\bar{z} + 3b_{12}z\bar{z}^2 + b_{03}\bar{z}^3 + \mathcal{O}_4 \\ \dot{\bar{z}} = -\lambda\bar{z} + \overline{a_{20}}z^2 + 2\overline{a_{11}}z\bar{z} + \overline{a_{02}}\bar{z}^2 + \overline{b_{30}}z^3 + 3\overline{b_{21}}z^2\bar{z} + 3\overline{b_{12}}z\bar{z}^2 + \overline{b_{03}}\bar{z}^3 + \mathcal{O}_4 \end{cases}$$

where

$$a \equiv a_{20} = a_{11} = a_{20} = \frac{\alpha}{8\lambda_c}\mathcal{I}_0'' - \frac{\alpha(\delta - \lambda_r)}{8\lambda_c^2}\mathcal{I}_0''\boldsymbol{i}$$

$$b \equiv b_{30} = b_{21} = b_{12} = b_{03} = \frac{\alpha}{48\lambda_c^2}\mathcal{I}_0''' - \frac{\alpha(\delta + \lambda_r)}{48\lambda_c^3}\mathcal{I}_0''\boldsymbol{i}\,,$$

Denoting

$$h_2(z,\bar{z}) = h_{220}z^2 + h_{211}z\bar{z} + h_{202}\bar{z}^2 \qquad\qquad z \to z + h_2(z,\bar{z})$$

$$\dot{z} = \lambda z + \left(-\lambda \frac{\partial h_2(z,\bar{z})}{\partial z}\, z + \lambda \frac{\partial h_2(z,\bar{z})}{\partial \bar{z}} \bar{z} + \lambda\, h_2(z,\bar{z}) + a_{20}\, z^2 + 2a_{11}\, z\bar{z} + a_{02}\, \bar{z}^2 \right) + \mathcal{O}_3$$

$$(-\lambda\, h_{200} + a_{20})\, z^2 + (3\lambda\, h_{202} + a_{02})\bar{z}^2 + (\lambda\, h_{211} + 2a_{11}) z\bar{z} = 0$$

$$\Leftrightarrow \qquad h_{220} = \frac{a_{20}}{\lambda}, \qquad h_{202} = -\frac{a_{02}}{3\lambda}, \qquad h_{221} = -\frac{2a_{11}}{\lambda}.$$

$$\dot{z} = \lambda z + f_{30}z^3 + f_{21}z^2\bar{z} + f_{12}z\bar{z}^2 + f_{03}\bar{z}^3 + \mathcal{O}_4$$

$$\begin{cases} f_{30} = b_{30} - 2\lambda h_{220}^2 - 2h_{220}a_{20}, \\ f_{21} = 3b_{21} - 5\lambda h_{220}h_{211} - h_{211}a_{20} - 4h_{220}a_{11}, \\ f_{12} = 3b_{12} - 6\lambda h_{220}h_{202} - 2\lambda h_{211}^2 - 2h_{211}a_{11} - 2h_{220}a_{02}, \\ f_{03} = b_{03} - 3\lambda h_{211}h_{202} - h_{211}a_{02}. \end{cases}$$

Repeating again with

$$h_3(z,\bar{z}) = h_{330}z^3 + h_{321}z^2\bar{z} + h_{312}z\bar{z}^2 + h_{303}\bar{z}^3 \qquad z \to z + h_3(z,\bar{z})$$

$$\dot{z} = \lambda z + \left(-\lambda \frac{\partial h_3(z,\bar{z})}{\partial z}\, z + \lambda \frac{\partial h_3(z,\bar{z})}{\partial \bar{z}}\, \bar{z} + \lambda\, h_3(z,\bar{z}) + f_{30}\, z^3 + f_{21}\, z^2\bar{z} + f_{12} z\bar{z}^2 + f_{03}\, \bar{z}^3 \right) + \mathcal{O}_4$$

$$(-2\lambda\, h_{330} z^2 + f_{30})\, z^3 + (4\lambda\, h_{303} + f_{03})\bar{z}^3 + f_{21}\, z^2\bar{z} + (2\lambda h_{312} + f_{12})\, z\bar{z}^2 = 0$$

$$\Leftrightarrow \qquad h_{330} = \frac{f_{30}}{2\lambda}, \qquad h_{303} = -\frac{f_{03}}{4\lambda}, \qquad h_{312} = -\frac{f_{12}}{2\lambda}.$$

$$\dot{z} = \lambda z + f_{21} z^2 \bar{z} + \mathcal{O}_4.$$

Recalling $z = \varphi + \phi \boldsymbol{i}$, we have

$$\boxed{\begin{cases} \dot{\varphi} = \lambda_r \varphi - \lambda_c \phi + (\nu\varphi - k\phi)(\varphi^2 + \phi^2) + \mathcal{O}_4, \\ \dot{\phi} = \lambda_r \phi + \lambda_c \varphi + +(\nu\varphi + k\phi)(\varphi^2 + \phi^2) + \mathcal{O}_4, \end{cases}}$$

where

$$\nu = \frac{\alpha \mathcal{I}_0'''}{16\lambda_c^2} + \frac{\alpha^2 \mathcal{I}_0''^2}{8|\lambda|^2 \lambda_c^2}\left(-2\delta - \lambda_r - \frac{(\delta + \lambda_r)^2 \lambda_r}{\lambda_c^2}\right),$$

$$k = -\frac{\alpha(\delta + \lambda_r)\mathcal{I}_0'''}{16\lambda_c^3} + \frac{\alpha^2 \mathcal{I}_0''^2}{4|\lambda|^2 \lambda_c}\left(-\frac{(\delta + \lambda_r)\lambda_r}{\lambda_c^2} + \frac{(\delta + \lambda_r)^2}{2\lambda_c^2} - \frac{1}{2}\right).$$

Polar coordinates

$\varphi = r\cos\theta\,, \phi = r\sin\theta \qquad r \in [0, +\infty) \quad \theta \in [0, 2\pi)$

Truncating the system at the third order

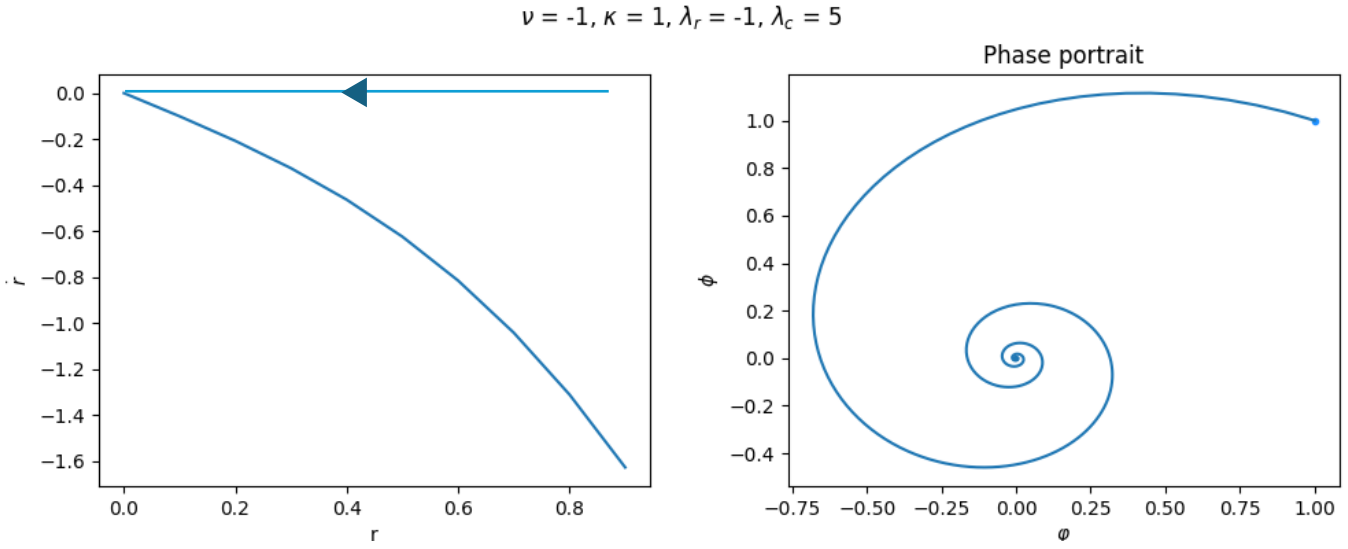
$$\begin{cases} \dot{r} = \lambda_r r + \nu r^3 \\ \dot{\theta} = \lambda_c + k r^2 \end{cases}$$

$$\dot{r} = 0$$

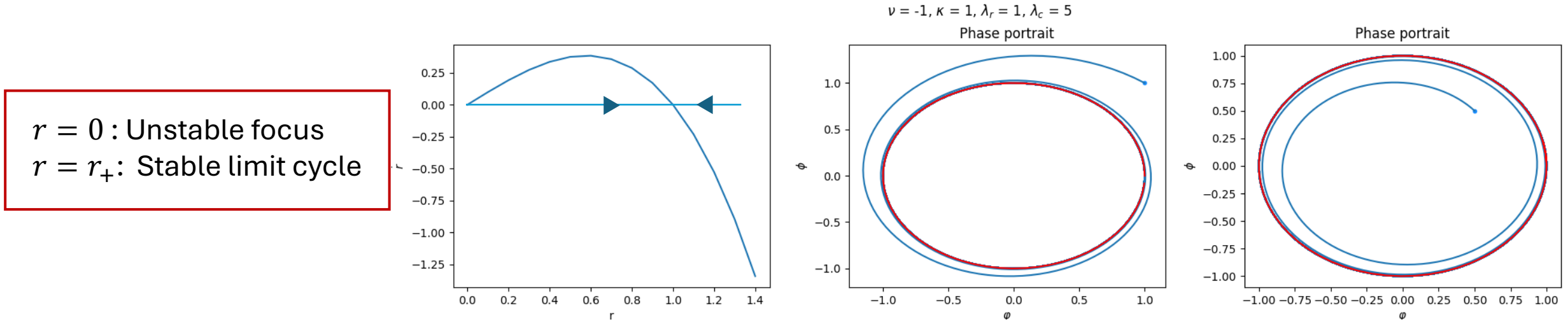
- $r = 0$ — If λ_r and ν share the same sign
- $r = 0,\ r = r_+ \equiv \sqrt{-\frac{\lambda_r}{\nu}}$ — If λ_r and ν do not share the same sign

$\nu < 0$

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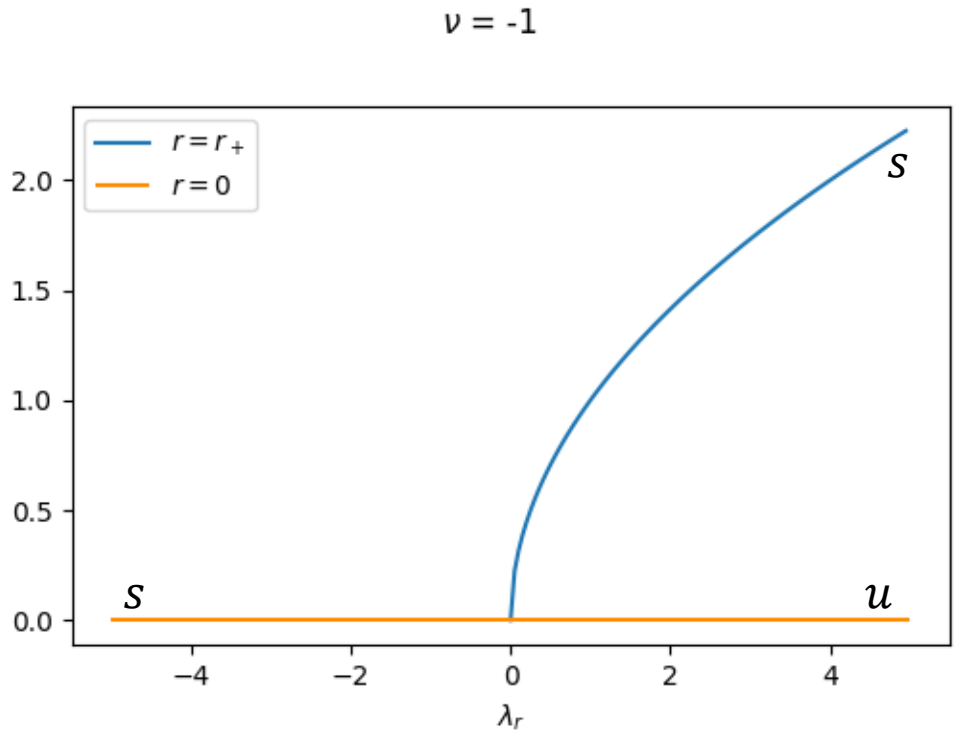
$r = 0$: Stable focus



$r = 0$: Unstable focus
$r = r_+$: Stable limit cycle

Bifurcation diagram

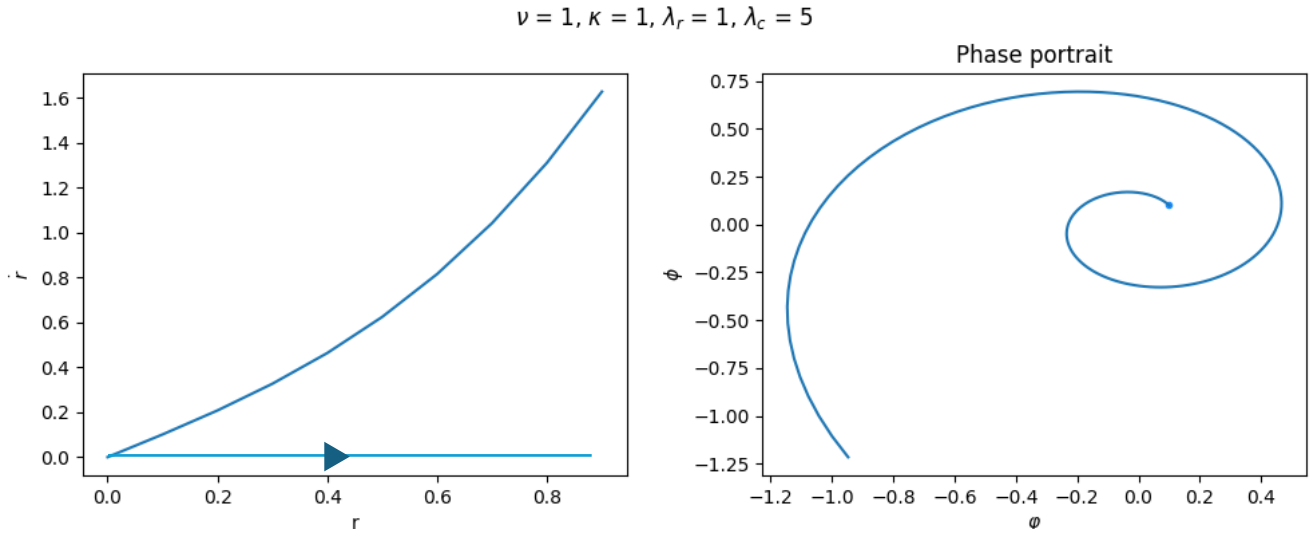
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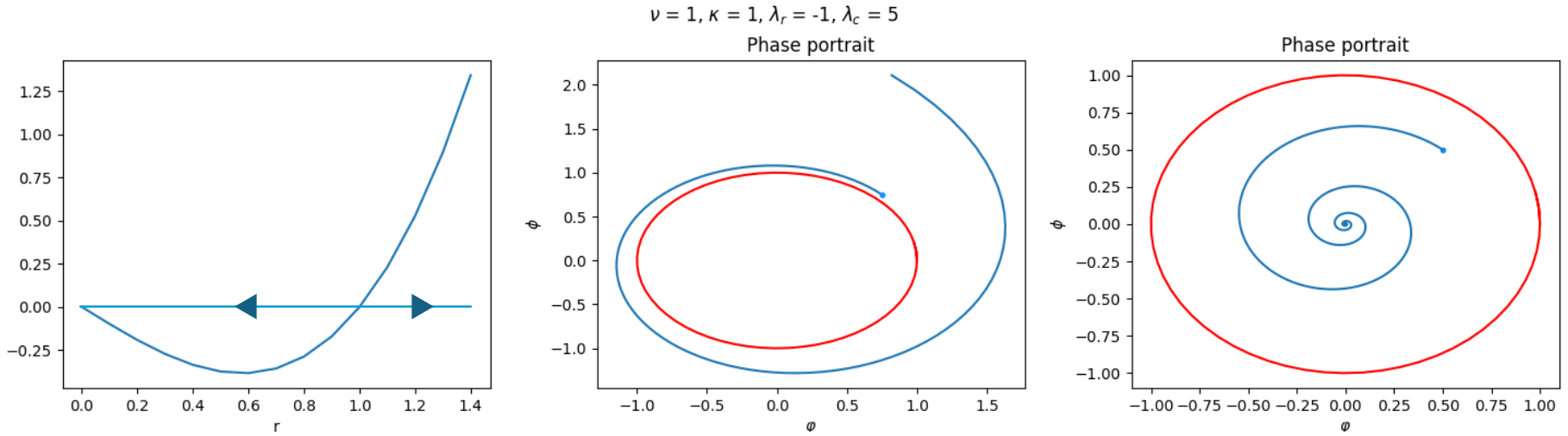
(Half) Pitchfork Bifurcation

$\nu > 0$

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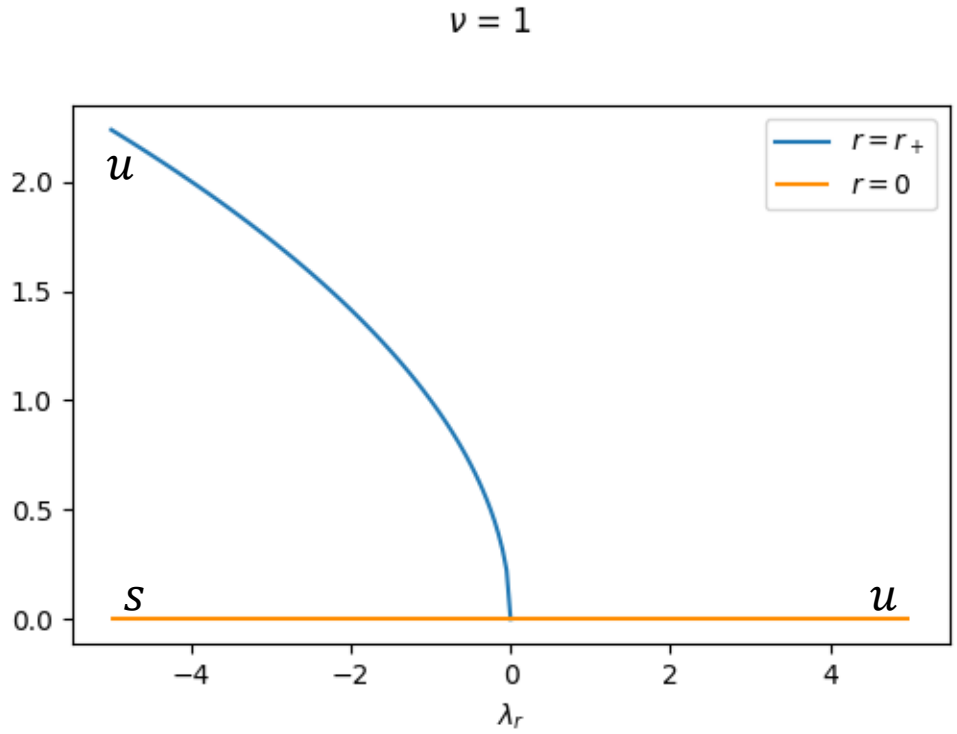
$r = 0$: Unstable focus



$r = 0$: Stable focus
$r = r_+$: Unstable limit cycle

Bifurcation diagram

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(Half) Pitchfork Bifurcation

Period Of The Limit Cycle

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On the limit cycle:

$$r = r_+ \implies \dot{\theta}(r_+) = \lambda_c - \frac{k\lambda_r}{\nu}$$

$$2\pi = \int_0^T dt\, \dot{\theta}(r_+)$$

$$T = \frac{2\pi\nu}{\nu\lambda_c - k\lambda_r}$$

Stochasticity

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Defining ξ_t as the white noise

$$\begin{cases} \dot{r} = \lambda_r r + \nu r^3 + \epsilon r \xi_t, \\ \dot{\theta} = \lambda_c + k r^2 \end{cases}$$

Stochastic term:

- ϵ "small" → system is robust
- As r increases, stochastic term increases
- $\langle \xi_t \rangle = 0 \quad \langle \xi_t \, \xi_{t'} \rangle = \delta(t - t')$

Stochasticity

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In terms of SDE:

$$\begin{cases} dr = (\lambda_r r + \nu r^3)dt + \epsilon r dB(t), \\ d\theta = (\lambda_c + kr^2)dt, \end{cases}$$

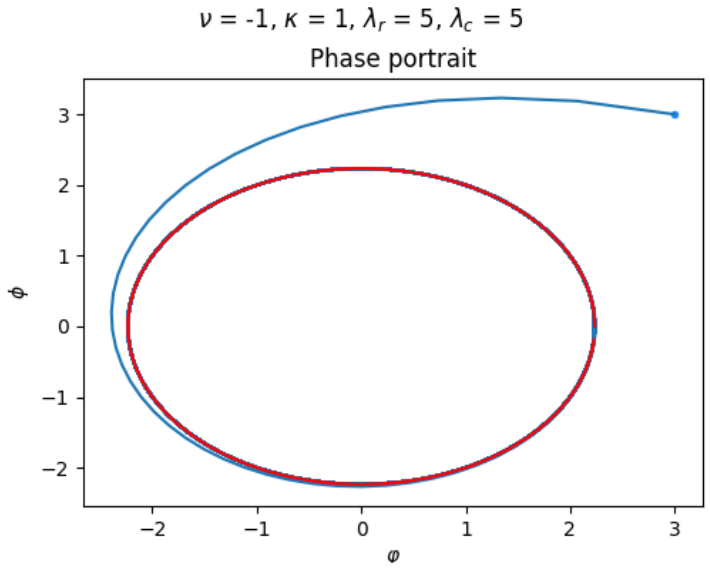
where $B(t)$ is a Brownian motion defined on a probability space $(\Omega, \mathcal{F})$:

- $B(0) = 0$
- $B(t) - B(s) \sim \mathcal{N}(0, t-s) \quad 0 \leq s < t$
- For all $0 \leq t_0 < t_1 < \cdots < t_n$, the increments $B(t_i) - B(t_{i-1})$ are indipendent

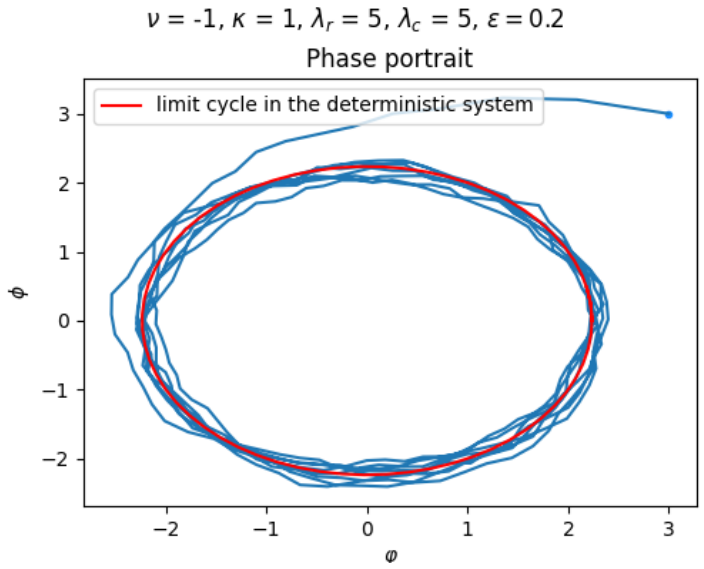
Simulations

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Numerical simulations using `sdeint` (python)



Adding noise



Using Itō prescription

$$dr = (\lambda_r\, r + \nu\, r^3)\, dt + \epsilon\, r\, dB(t).$$

Substituting $x = r^{-2}$

$$dx = \frac{\partial x}{\partial r}\, dr + \frac{1}{2}\frac{\partial^2 x}{\partial r^2}\, (dr)^2 + \cdots$$

$$(dr)^2 = \epsilon^2 r^2\, dt + \cdots, \qquad \big(dB(t)\big)^2 = dt$$

$$dx = (-2\lambda_r x - 2\nu + 3\epsilon^2 x)\, dt - 2\epsilon\, x\, dB(t)$$

Homogeneous Equation

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Homogeneous equation

$$dx_h = -(2\lambda_r - 3\epsilon^2)\, x_h\, dt - 2\epsilon\, x_h\, dB(t)$$

Substituting $\mathrm{y} = \ln x_h$

$$dy = -(2\lambda_r - \epsilon^2)\, dt \; - 2\epsilon\, dB(t)$$

$$d \ln x_h = -\,(2\lambda_r - \epsilon^2)\, dt \; - 2\epsilon\, dB(t)$$

$$\boxed{x_h(t) = x_h(t_0)\, e^{-\left(2\lambda_r - \epsilon^2\right)(t - t_0) - 2\epsilon[B(t) - B(t_0)]}}$$

By variation of constants:

$$x(t) = c(t)\, e^{-\left(2\lambda_r - \epsilon^2\right)(t-t_0) - 2\epsilon[B(t)-B(t_0)]} \qquad c(t_0) = x(t_0)$$

By comparison with the original equation in x:

$$dx(t) = x_h(t)\, dc(t) - (2\lambda_r - 2\epsilon^2)x(t)dt \; - 2\epsilon x(t)dB(t)$$

$$c(t) = x(t_0) - 2\nu \int_{t_0}^{t} e^{\left(2\lambda_r - \epsilon^2\right)(s-t_0) + 2\epsilon[B(s)-B(t_0)]}$$

Pullback Attractor

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$$x(t) = \left(x(t_0) - 2\nu \int_{t_0}^{t} ds\, e^{\left(2\lambda_r - \epsilon^2\right) s + 2\epsilon B(s)} \right) e^{-\left(2\lambda_r - \epsilon^2\right) t - 2\epsilon B(t)}$$

In deterministic systems $\longrightarrow$ $t_0 = 0,\ t \to +\infty,$

In stochastic systems $\longrightarrow$ $t_0 \to -\infty,\ t = 0$ (Arnold)

$$\lim_{t_0 \to -\infty} x(t_0) = \lim_{t_0 \to -\infty} \frac{1}{r^2(t_0)} = 0 \qquad x(0) = r_s^{-2} \text{ (stationary random radius)}$$

$$r_s = \left(-2\nu \int_{-\infty}^{0} ds\, e^{\left(2\lambda_r - \epsilon^2\right) s + 2\epsilon B(s)} \right)^{-1/2}$$

Forward Fokker Plank Equation

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$$dX(t) = \mu(x,t)\,dt + \sigma(x,t)\,dB(t)$$

$p(x,t)$: probability that a stochastich trajectory $X(t)$ falls in $[x, x+dx)$ at time t

Forward Fokker Plank equation

$$\frac{\partial p(x,t)}{\partial t} = -\frac{\partial}{\partial x} J(x,t)$$

with

$$J(x,t) = \mu(x,t)\,p(x,t) - \frac{1}{2}\frac{\partial}{\partial x}[\sigma^2(x,t)p(x,t)]$$

In our case

$$J(r,t) = (\lambda_r r + \nu r^3)p(r,t) - \frac{1}{2}\frac{\partial}{\partial x}[\epsilon^2 r^2 p(r,t)]$$

Equilibrium distribution

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Equilibrium condition: $J = 0$

$$(\lambda_r r + \nu r^3)\, p_{eq}(r) = \frac{\partial}{\partial r}\left[\frac{1}{2}\epsilon^2 r^2 p_{eq}(r)\right],$$

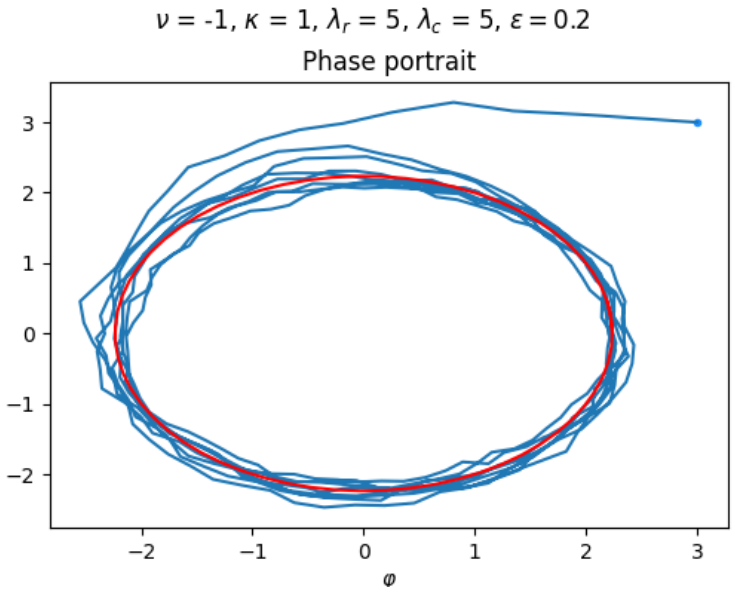
$$p_{eq}(r) = C\, r^{\frac{2(\lambda_r - \epsilon^2)}{\epsilon^2}}\, e^{\frac{\nu}{\epsilon^2} r^2}$$

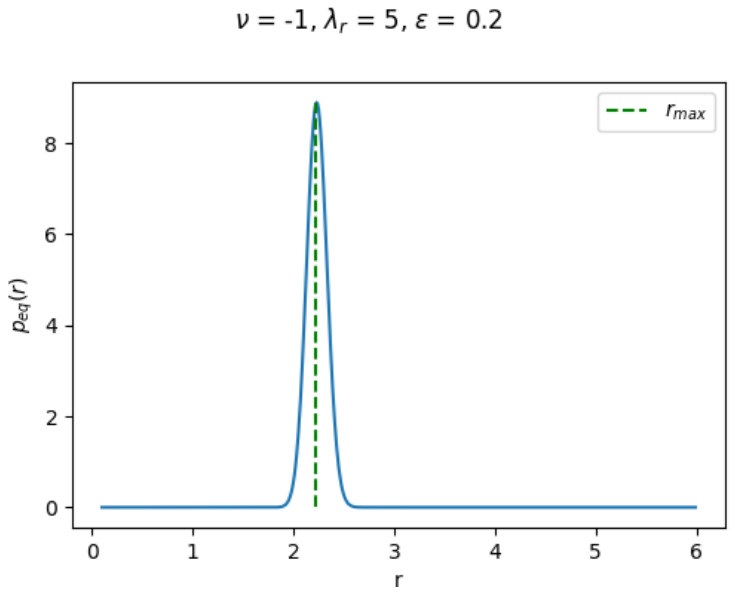
Normalisation condition

$$\int_0^{+\infty} dr\, C\, r^{\frac{2(\lambda_r - \epsilon^2)}{\epsilon^2}}\, e^{\frac{\nu}{\epsilon^2} r^2} = 1$$

Note: $\nu < 0 \Longrightarrow$ convergence of the integral

$$p_{eq}(r) = 2\left(-\frac{\epsilon^2}{\nu}\right)^{\frac{1}{2}-\frac{\lambda_r}{\epsilon^2}} \Gamma^{-1}\left(\frac{\lambda_r}{\epsilon^2}-\frac{1}{2}\right) r^{\frac{2(\lambda_r-\epsilon^2)}{\epsilon^2}} e^{\frac{\nu}{\epsilon^2}r^2}$$





Expectation Values

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Most likely radius (mode of the stationary distribution):

$$\frac{dp_{eq}}{dr}(r_{max}) = 0 \qquad \Rightarrow \qquad r_{max} = \sqrt{-\frac{\lambda_r - \epsilon^2}{\nu}} \qquad (\epsilon \ll \lambda_r)$$

Mean stationary radius (average over all possible realizations of the noise):

$$\mathbb{E}[r_s] = \int_0^{+\infty} dr\, r\, p_{eq}(r) = \sqrt{-\frac{\epsilon^2}{\nu}}\,\Gamma\left(\frac{\lambda_r}{\epsilon^2}\right)\Gamma^{-1}\left(\frac{\lambda_r}{\epsilon^2} - \frac{1}{2}\right)$$

In the limit $\epsilon \to 0$ (no noise):

$$p_{eq}(r) \to \delta(r - r_+) \qquad r_{max}, \mathbb{E}[r_s] \to r_+ \qquad r_+ = \sqrt{-\frac{\lambda_r}{\nu}}$$

$$\mathbb{E}[r_s^2] = \int_0^{+\infty} dr\, r^2\, p_{eq}(r) = -\frac{\epsilon^2}{\nu}\Gamma\left(\frac{\lambda_r}{\epsilon^2}+\frac{1}{2}\right)\Gamma^{-1}\left(\frac{\lambda_r}{\epsilon^2}-\frac{1}{2}\right)$$

Variance (half thickness of the cloud):

$$\sigma = \sqrt{\mathbb{E}[r_s^2] - \mathbb{E}[r_s]}$$

Period Of The Limit Cycle

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$$2\pi = \int_0^{T(r_s)} dt\, \dot{\theta}(r_s)$$

$$\mathbb{E}[\dot{\theta}(r_s)] = \lambda_c + k\, \mathbb{E}[r_s^2] = \lambda_c - \frac{k\epsilon^2}{\nu} \Gamma\left(\frac{\lambda_r}{\epsilon^2} + \frac{1}{2}\right) \Gamma^{-1}\left(\frac{\lambda_r}{\epsilon^2} - \frac{1}{2}\right)$$

$$\mathbb{E}[T(r_s)] = \frac{2\pi\, \nu\, \Gamma\left(\frac{\lambda_r}{\epsilon^2} - \frac{1}{2}\right)}{\nu\lambda_c \Gamma\left(\frac{\lambda_r}{\epsilon^2} - \frac{1}{2}\right) - k\epsilon^2 \Gamma\left(\frac{\lambda_r}{\epsilon^2} + \frac{1}{2}\right)}$$

Period of the most likely limit cycle

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$$2\pi = \int_0^{T(r_{max})} dt\, \dot{\theta}(r_{max})$$

$$T(r_{max}) = \frac{2\pi\nu}{\lambda_c\nu - k\lambda_r + k\epsilon^2}$$

In the limit $\epsilon \to 0$:

$$\mathbb{E}[T(r_s)], T(r_{max}) \to T \qquad T = \frac{2\pi\nu}{\lambda_c\nu - k\lambda_r}$$

Conclusions

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$$Y, K \longrightarrow \varphi, \phi \longrightarrow z, \bar{z} \longrightarrow r, \theta$$

$\lambda_r < 0 \longrightarrow \lambda_r > 0,$ (Half) Pitchfork Bifurcation

$\nu < 0,\ \lambda_r > 0$: Stable Limit Cycle

$$r_+ = \sqrt{-\frac{\lambda_r}{\nu}}\,,$$

$$T = \frac{2\pi\nu}{\nu\lambda_c - k\lambda_r}$$

$$\mathbb{E}[r_s] = \sqrt{-\frac{\epsilon^2}{\nu}}\,\Gamma\left(\frac{\lambda_r}{\epsilon^2}\right)\Gamma^{-1}\left(\frac{\lambda_r}{\epsilon^2} - \frac{1}{2}\right), r_{max} = \sqrt{-\frac{\lambda_r - \epsilon^2}{\nu}}$$

$$\mathbb{E}[T(r_s)] = \frac{2\pi\,\nu\,\Gamma\left(\frac{\lambda_r}{\epsilon^2} - \frac{1}{2}\right)}{\nu\lambda_c\Gamma\left(\frac{\lambda_r}{\epsilon^2} - \frac{1}{2}\right) - k\epsilon^2\Gamma\left(\frac{\lambda_r}{\epsilon^2} + \frac{1}{2}\right)}$$

$$T(r_{max}) = \frac{2\pi\nu}{\lambda_c\nu - k\lambda_r + k\epsilon^2}$$

Thanks for the attention

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